

ScottDomains

A Lean 4 / Mathlib formalization of **domain theory**, developed directly from one paper:

C. A. Gunter and D. S. Scott, “**Semantic Domains**,” in *Handbook of Theoretical Computer Science*, Volume B (ed. J. van Leeuwen), North-Holland, 1990, pp. 633–674.

That paper is our **specification**. We work through its development in order —

1. §2 cpo’s and the Fixed-Point Theorem
2. §3 effectively presented domains (compact elements, algebraic, bounded-complete, finitary projections)
3. §4 operators and functions (products, smash products, sums, lifts, function spaces)
4. §5 powerdomains
5. §6 bifinite (SFP) domains
6. §7 recursive definitions of domains (D^∞)

— formalizing each definition and theorem in Lean, **reusing Mathlib** where it already provides a piece and **defining the rest** (notably the way-below relation $\ll$, which Mathlib lacks).

The paper — a copy provided by **D. Scott** himself, whose distribution to us is permitted under the publisher’s author terms — is in [papers/Gunter Scott 1990.pdf](#).

Layout

- [ScottDomains.lean](#) — root module; re-exports the Mathlib foundations we build on.
- [ScottDomains/ExistingTheories.lean](#) — clickable #check catalog of the Mathlib pieces we reuse.
- [docs/PaperInventory.md](#) — every definition and theorem in the paper, with a column for its Lean/Mathlib status (the work list).
- [papers/](#) — the source paper.

Status

Framing complete: the project builds green (`lake build` → 924 jobs, 0 errors, 0 sorry). The domain-theory development itself — the classes, the $\ll$ relation, and the theorems — is **not yet started**. `docs/PaperInventory.md` is the plan, mapped section-by-section to the paper; the next step is to fill in its “In Lean / Mathlib?” column, then formalize.